



Security and Cyber Laws

Block-1 Unit-1

Cyber Security Issues and Challenges

Topics to be Covered

- Introduction
- Digital Security
- Digital Security Pros
- Digital Security Cons
- Security Issues/Breaches in Cyberspace
- Cyber Security and Law
- Important Topics
- Summary

Introduction

- Cyber security is the protection of Internet-connected systems, including hardware, software, and data from cyber attacks.
- It is made up of two words one is cyber and other is security.
- Cyber is related to the technology which contains systems, network and programs or data.
- The security is related to the protection which includes systems security, network security and application and information security.
- It is the body of technologies, processes, and practices designed to protect networks, devices, programs, and data from attack, theft, damage, modification or unauthorized access. It may also be referred to as information technology security.

Digital Security

- Digital security means protecting your computer, mobile devices, tablets, and any other Internet-connected devices from intruders, which could be in the form of hacking, phishing, and more.
- Digital security could also be used to protect your personal data from being used and sold by companies.
- There are a number of ways to protect yourself online, from VPNs to password managers and identity monitoring services.



www.istockphoto.com/photos/digital-security

Digital Security



- The 3 pillars of digital security are:
 - Confidentiality
 - Integrity
 - Availability
- The basic essence of these principles is that the information which is private should be shared with the least amount of people to keep it more secure.
- The information provided should not be modified or corrupted.
- The information provided should work effectively and efficiently at all times.

Digital Security: Pros

- It helps in protecting personal information stored in devices.
- Suspicious or unauthorized access to devices can be blocked through digital security.
- Secure the hacking method.
- Provide confidentiality of data and system. This can be achieved by enforcing security policies and network protocols well.
- Protects the computer from crashing or slowing down and thus protects business, transactions, communication etc. happening over computer, network or system.
- Digital security thus may help in fostering the economy of the State as it cuts down on many costs.

Digital Security: Cons

- Availing services or procuring tools for digital security can be a costly affair.
- Web services or tools may or may not be compatible with the device of the user.
- Digital security services or tools may be difficult to configure at times and needs to be updated regularly.
- Services or tools may slow down functioning of user's device or at times may intervene even normal functioning of another programme.

Security Issues /Breaches in Cyberspace

- Cyberspace refers to the virtual computer world, and more specifically, an electronic medium that is used to facilitate online communication.
- Cyberspace typically involves a large computer network made up of many worldwide computer sub networks that employ TCP / IP protocol to aid in communication and data exchange activities.
- Cyberspace allows users to share information, interact, swap ideas, play games, engage in discussions or social forums, conduct business and create intuitive media, among many other activities.

Security Issues /Breaches in Cyberspace

- **Unauthorized access**— It is accessing computer, computer, network, system or device without permission from those who are authorized to access the same.
- **Distributed Denial of Service Attack**
 - A distributed denial of service (DDoS) attack is when an attacker, or attackers, attempt to make it impossible for a service to be delivered.
 - This can be achieved by thwarting access to virtually anything: servers, devices, services, networks, applications, and even specific transactions within applications.
 - It's one system that is sending the malicious data or requests; a DDoS attack comes from multiple systems.

Security Issues /Breaches in Cyberspace

- **Malware**, or malicious software, is any program or file that is harmful to a computer user.
- Types of malware can include **computer viruses, worms, Trojan horses and spyware**.
- These malicious programs can perform a variety of functions such as stealing, encrypting or deleting sensitive data, altering or hijacking core computing functions and monitoring users' computer activity.

Security Issues /Breaches in Cyberspace

- **Social engineering attacks**
 - It involves some form of psychological manipulation, fooling otherwise unsuspecting users or employees into handing over confidential or sensitive data.
 - Commonly, social engineering involves email or other communication that invokes urgency, fear, or similar emotions in the victim.



Source : <https://www.shutterstock.com/>

Security Issues /Breaches in Cyberspace

- **Phishing**

- Phishing attacks attempt to gain sensitive, confidential information such as usernames, passwords, credit card information, network credentials, and more.

- **Crypto Jacking**

- It is a type of cybercrime that involves the unauthorized use of people's devices (computers, smartphones, tablets, or even servers) by cybercriminals to mine for crypto currency.
- Crypto currency is a virtual currency with no central regulatory or issuing authority.

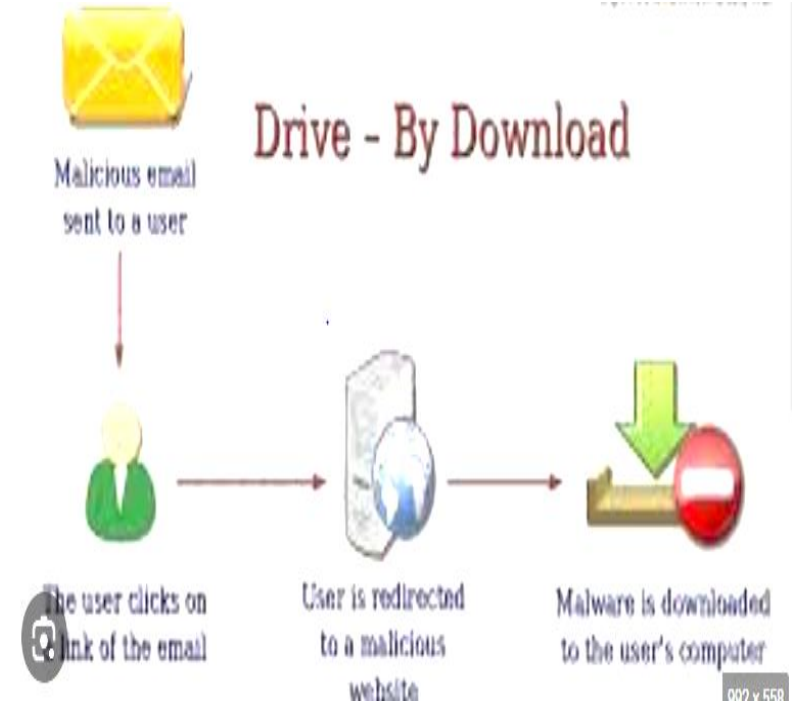
Security Issues /Breaches in Cyberspace

- **Exploiting vulnerability**
 - Vulnerability is a flaw in the measures taken to secure a device. Such flaw is exploited by the attacker to gain access over the device or harm it.
- **Cyber physical attacks**
 - These are cyber-attacks breaching the security and impacting the physical environment. This may include shutting down of cameras, lights etc. which are cyber controlled.

Security Issues /Breaches in Cyberspace

- **Drive by download**

- Drive by download attacks specifically refer to malicious programs that install to your devices - without your consent.
- This also includes unintentional downloads of any files or bundled software onto a computer device.



Some Common Security Issues

- **Internet time theft**
 - This involves usage by unapproved persons of the internet hours paid for by another person.
- **Key Loggers**
 - It is a software database or a program intended to covertly keep an eye on and log all keystrokes. The Key logger software scans computers, their processes, and data, the moment a person hits a key on the keyboard.



<https://www.istockphoto.com/photos/cyber-security>

Some Common Security Issues

- **Website defacement**

- This is usually transmitted by the replacement of the homepage of a site by a system cracker that dislocated into a web server and modifies the hosted website creating one of its own.
- The attacker usually replaces the site matters with his own message or completely destroys the site's contents.



<https://www.malcare.com/blog/website-defacement-attack/>

Some Common Security Issues

- **Pharming**

- This takes place when the attacker redirects a user from an authentic and genuine site to a fake and deceitful site where their systems are infused with malware.

- **Phreaking**

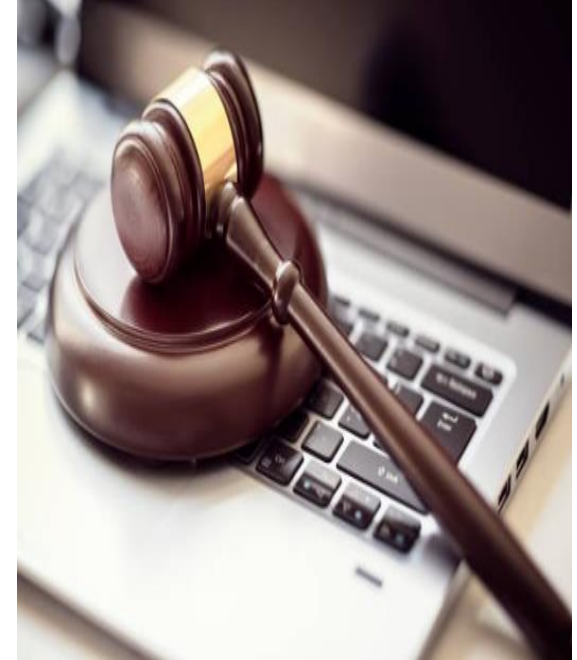
- This refers to people who interfere with systems of telecommunication such as public networks.

- **Email bombing**

- This refers to forwarding a significant large number of emails to the victim resulting in the victim's email account or mail servers to not respond.

Cyber Security and Law

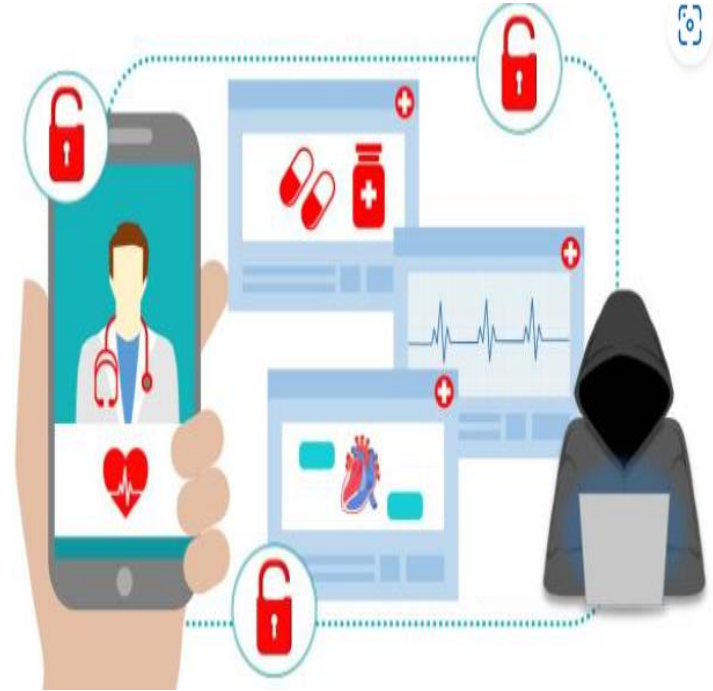
- Cyber Law has played an instrumental role in regulating security issues and breaches in cyberspace.
- Cyber law consists of Acts, Rules, Regulations, Notifications etc. passed by the Government of India.
- Cyber security law is concerned with integrity, confidentiality, availability of public private information systems and seeks to protect individual rights like privacy, economic interests and national security.



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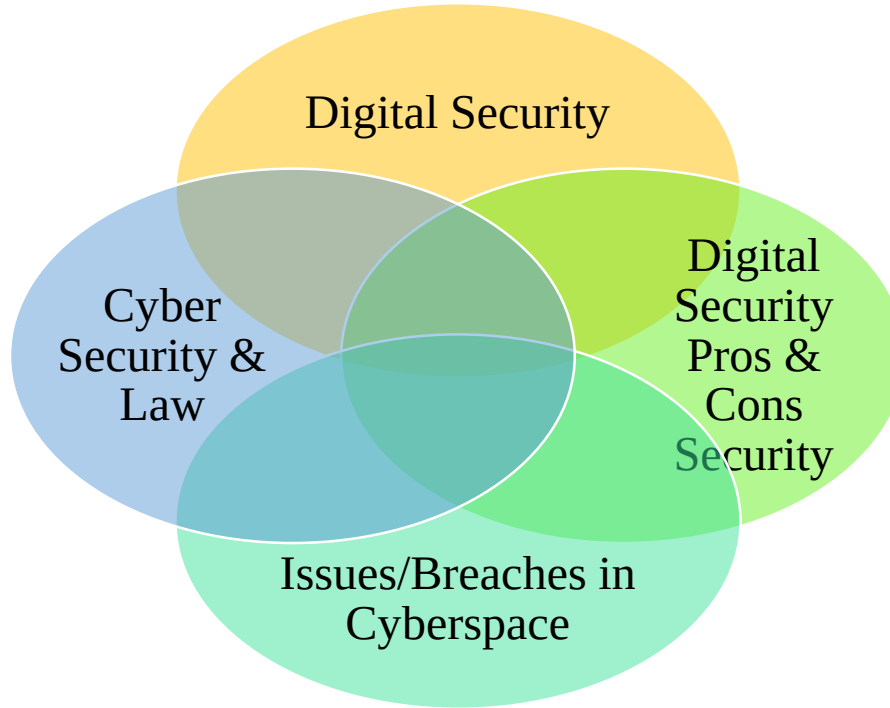
Cyber Security and Law

- Unauthorized access
- Distributed Denial of Service Attack
- Malwares
- Social Engineering attacks
- Crypto jacking
- Exploiting vulnerability
- Cyber physical attacks
- IOT attacks
- Web Jacking
- Drive by download



<https://www.istockphoto.com/vector/cyber-security-threat-at-online-protected-healthcare-information-database>

Important Topics



Summary

- In this session, we learnt about challenges to cyber security exists in the form of unauthorized access, exploiting vulnerability, cyber physical attacks, IOT attacks, Web jacking, drive by download, crypto jacking, social engineering attack, malwares, denial of service attacks etc.
- Cyber security attacks may also cause intellectual property violations and hence attract law regulating the same.
- An information technology law empowers enforcement agencies with investigative powers and recognizes electronic evidence admissible in the courts of law.

*Thank
You*



Security and Cyber Laws

Block-1 Unit-2 (Part-2)

Cryptography Mechanism

Topics to be Covered

- Commonly used crypto algorithm
 - - DES
 - - RSA
- Electronic Signature
- Authentication and Authorization
 - - Hash Functions
 - - Access Control Derivatives
- Key establishment, management and certification
- Trusted third parties and public key certificates

Topics to be Covered

- Pseudorandom numbers and sequence
- Public Key infrastructure
- Data Encryption standards
- Important Topics
- Summary

Introduction

- **Data encryption** translates data into another form, or code, so that only people with access to a secret key (formally called a decryption key) or password can read it.
- Encrypted data is commonly referred to as ciphertext, while unencrypted data is called plaintext. Currently, encryption is one of the most popular and effective data security methods used by organizations.



Commonly used Crypto Algorithms

- **Data Encryption Standard (DES):**
 - It was originally developed by IBM and was adopted as an NBS Standard in 1977.
 - DES is a symmetric cryptosystem, so the cipher text is decrypted using the same key.
 - It operates on 64-bit blocks of input at a time.
 - DES uses a 56-bit encryption key, and it's based on the Feistel Structure that was designed by a cryptographer named Horst Feistel.
 - DES converts 64-bit blocks of plaintext data into cipher text by dividing the block into two separate 32-bit blocks and applying the encryption process to each independently.

Commonly used Crypto Algorithms

- **Steps of the DES encryption algorithm operating on 64-bit block are:**
 - The process begins with the 64-bit plain text block getting handed over to an initial permutation (IP) function.
 - The initial permutation (IP) is then performed on the plain text.
 - Next, the initial permutation (IP) creates two halves of the permuted block, referred to as Left Plain Text (LPT) and Right Plain Text (RPT).
 - Each LPT and RPT goes through 16 rounds of the encryption process.
 - Finally, the LPT and RPT are rejoined, and a Final Permutation (FP) is performed on the newly combined block.
 - The result of this process produces the desired 64-bit ciphertext.

Commonly used Crypto Algorithms

- **RSA Algorithm:**
 - The RSA algorithm is an asymmetric cryptography algorithm; this means that it uses a public key and a private key, a public key is shared publicly, while a private key is secret and must not be shared with anyone.
 - They all produced this algorithm, which is totally based on modular mathematics of Number theory. It is an asymmetric cryptography algorithm because it uses two different keys for encoding and decoding.
 - The RSA algorithm is named after those who invented it in 1978: Ron Rivest, Adi Shamir, and Leonard Adleman.

Commonly used Crypto Algorithms

- **RSA Algorithm:**
- **Encryption**
 - Given a plaintext PP , represented as a number, the cipher text CC is calculated as: $C = P^e \bmod n$
- **Decryption**
 - Using the private key $(n,d)(n,d)$, the plaintext can be found using:
 - $P = C^d \bmod n$.
- **Hybrid Encryption is the best of both symmetric and asymmetric encryption**, symmetric encryption method is fast for large data encryption but falls short of identity verification. The asymmetric encryption method is quite opposite, it is slow but have public and private key pair which is a must when it comes to cyber safety.

Electronic Signature

- A Digital Signature is a technique by which it is possible to secure electronic information in such a way that the originator of the information, as well as the integrity of the information, can be verified.
- This procedure of guaranteeing the origin and the integrity of the information is also called Authentication.
- The authenticity of many legal, financial, and other documents is determined by the presence or absence of an authorized handwritten signature.
- For a computerized message system to replace the physical transport of paper and ink documents handwritten signatures have to be replaced by Digital Signatures.

Electronic Signature

- A system by which one party can send a “signed” message to another party in such a way that
- The receiver can verify the claimed identity of the sender.
- The sender cannot repudiate the contents of the message.
- The receiver cannot possibly have concocted the message himself/herself.



Comparison between Electronic Signature and Digital Signature

- **Electronic Signature:** “Authentication of any electronic record by a subscriber by means of the electronic technique specified in the second schedule and includes digital signature.”
e.g. email signatures, web based signatures, digitized image of a signature.
- Electronic Signatures has no expiry or validity period unlike a digital signature which is valid up to a maximum of three years. Electronic Signatures are used for verifying a document unlike digital signatures which is used for securing a document.

Comparison between Electronic Signature and Digital Signature

- **Digital signatures** : “Authentication of any electronic record by a subscriber by means of the electronic technique specified in the second schedule and includes digital signature.”
e.g. signing tenders, bidding, e-filing for ROC or income tax or for GST.
- It follows an approach of using hash function i.e., usage of private key and a public key which is a two way protection system.

Source :SIGNiX



Authentication and Authorization

- Authentication is a term which is used in a very broad sense.
- Authentication is the process of identifying users and validating who they claim to be.
- One of the most common and obvious factors to authenticate identity is a password. If the user name matches the password credential, it means the identity is valid, and the system grants access to the user.



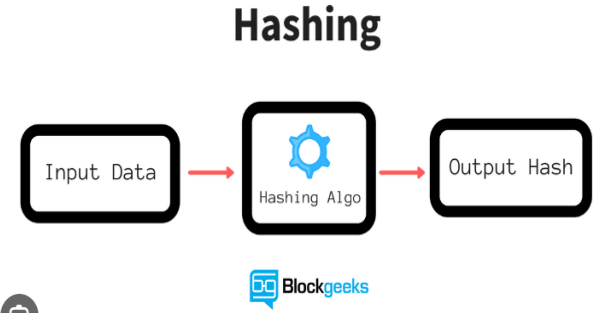
Blog post: Authentication and Digital Signatures
a 2-in-1 Approach

Authentication and Authorization

- Authorization occurs after a user's identity has been successfully authenticated.
- It is about offering full or partial access rights to resources like database, funds, and other critical information to get the job done.
- Authentication is done prior to authorization process.
- The process of authentication requires user's login details while authorization only requires user's privilege or security levels.

Hash Functions

- Hash functions are extremely useful and appear in almost all information security applications.
- A hash function is a mathematical function that converts a numerical input value into another compressed numerical value. The input to the hash function is of arbitrary length but output is always of fixed length.
- Values returned by a hash function are called message digest or simply hash values.



Hash Functions

- The most common cryptographic uses of hash functions are with digital signatures and for data integrity. With digital signatures, a long message is usually hashed and only the hash-value is signed.
- The party receiving the message then hashes the received message and verifies that the received signature is correct for this hash-value.
- This saves both time and space compared to signing the message directly.
- Hash functions may be used for data integrity, The hash-value corresponding to a particular input is computed at some point in time. The integrity of this hash-value is protected, to verify that the input data has not been altered.
- A third application of hash functions is their use in protocols involving prior commitments, including some digital signature schemes and identification protocols.

Hash Functions

- **Some of the popular hash functions include:-**
- The Message Digest family comprises of hash functions MD2, MD4, MD5 and MD6. It was adopted as Internet Standard RFC 1321. It is a 128-bit hash function.
- Secure Hash Functions family consists of four SHA algorithms; SHA-0, SHA-1, SHA-2, and SHA-3.
- The RIPEMD is an acronym for RACE Integrity Primitives Evaluation Message Digest. This set of hash functions was devised by open research community and generally known as a family of European hash functions.
- Whirlpool is a 512-bit hash function which has stemmed from the revised edition of Advanced Encryption Standard (AES). One of the designer was Vincent Rijmen, a co-creator of the AES.

Access Control Derivatives/ Mechanisms

- A cryptographic protocol (protocol) is a distributed algorithm defined by a sequence of steps precisely specifying the actions required of two or more entities to achieve a specific security objective.
- Protocols play a major role in cryptography and are essential in meeting cryptographic goals.
- Encryption schemes, digital signatures, hash functions, and random number generation are among the primitives which may be utilized to build a protocol.

Key Establishment Management and Certification

- **Key** establishment is any process whereby a shared secret key becomes available to two or more parties, for subsequent cryptographic use.
- **Key management** is the set of processes and mechanisms which support key establishment and the maintenance of ongoing keying relationships between authorized parties, including replacing older keys with new keys as and when necessary.

Key Establishment Management and Certification

- Key establishment can be broadly subdivided into
 - **key agreement**
 - **key transport**
- Key management encompasses techniques and procedures supporting:
 - initialization of system users within a domain;
 - generation, distribution, and installation of keying material;
 - controlling the use of keying material;
 - update, revocation, and destruction of keying material; and
 - Storage, backup/recovery, and archival of keying material.

Key Establishment Management and Certification

- Key management through symmetric-key techniques
- A symmetric cryptographic system is a system involving two transformations
 - one for the originator
 - and one for the recipient
- Both of which make use of either the same secret key (symmetric key) or two keys easily computed from each other.
- An asymmetric cryptographic system is a system involving two related transformations
- One is defined by a public key (the public transformation), and another defined by a private key (the private transformation).

Key Establishment Management and Certification

- **Advantages**

- It is easy to add and remove entities from the network.
- Each entity needs to store only one long-term secret key.

- **Disadvantages**

- All communications require initial interaction with the TTP.
- The TTP must store long-term secret keys.
- The TTP has the ability to read all messages.
- If the TTP is compromised, all communications are insecure.

Key Establishment Management and Certification

- **Key management through public-key techniques**
- There are a number of ways to address the key management problem through public-key techniques.
- Each entity in the network has a public/private encryption key pair. The public key along with the Identity of the entity is stored in a central repository called a public file.
- **Advantages**
- No trusted third party is required.
- The public file could reside with each entity.
- Only public keys need to be stored to allow secure communications between any pair of entities.

Trusted Third Party and Public Key Certification

- A trusted third party (TTP) is an entity in a domain that is trusted to perform a specific service.
- Various third party services require different types of trust and competency in the third party.
- An example, a third party possessing secret decryption keys (or entity authentication keys) must be trusted not to disclose encrypted information.
- **Definition A TTP is said to be functionally trusted if the entity is assumed to be honest and fair but it does not have access to the secret or private keys of users.**

Trusted Third Party and Public Key Certification

- **Public-key certificates**
- Public keys is generally easier than that of symmetric keys.
- Primary advantages offered by public-key (vs symmetric-key) techniques for applications related to key management includes:
- Simplified key management
- On-line trusted server not required.
- Enhanced functionality.

Trusted Third Party and Public Key Certification

- A public-key certificate consists of a data part and a signature part.
- The data part consists of the name of an entity, the public key corresponding to that entity, possibly additional relevant information (e.g. The entity's street or network address, a validity period or the public key, and various other attributes).
- The signature part consists of the signature of a TTP over the data part.

Source : Wikipedia.org



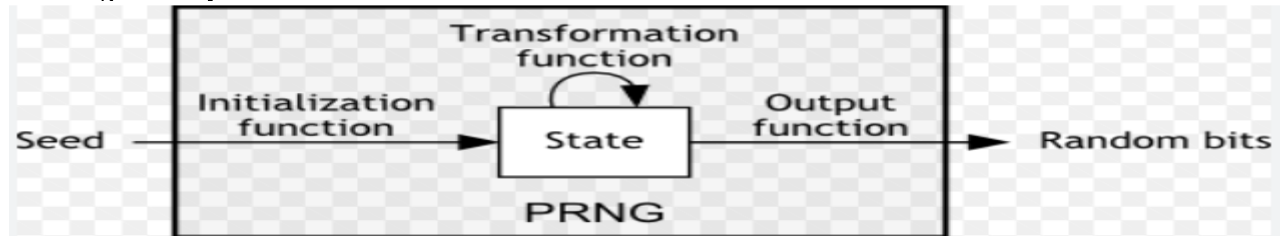
Pseudorandom Numbers And Sequences

- Pseudo Random Number Generator (PRNG) refers to an algorithm that uses mathematical formulas to produce sequences of random numbers.
- For example, keys for encryption transformations need to be generated in a manner which is unpredictable to an adversary.
- In cryptographic applications, one of the following steps must be performed:
 - From a finite set of elements, select an element at random.
 - From the set of all sequences (strings) of length over some finite alphabet of symbols, select a sequence at random.
 - Generate a random sequence (string) of symbols of length over a set of symbols.

Pseudorandom Numbers And Sequences

- It refers to an algorithm that utilizes mathematical formulation to create series of random numbers.
- They produce a series of numbers assessing the properties of random numbers.
- Some of the advantages are that this system is **efficient** as it can create number in a short span of time, easy to **determine** if replaying the same sequence of numbers again at a later stage and lastly are **periodic** i.e., that the sequence will eventually repeat itself.

Source : <https://pit-claudel.fr/>



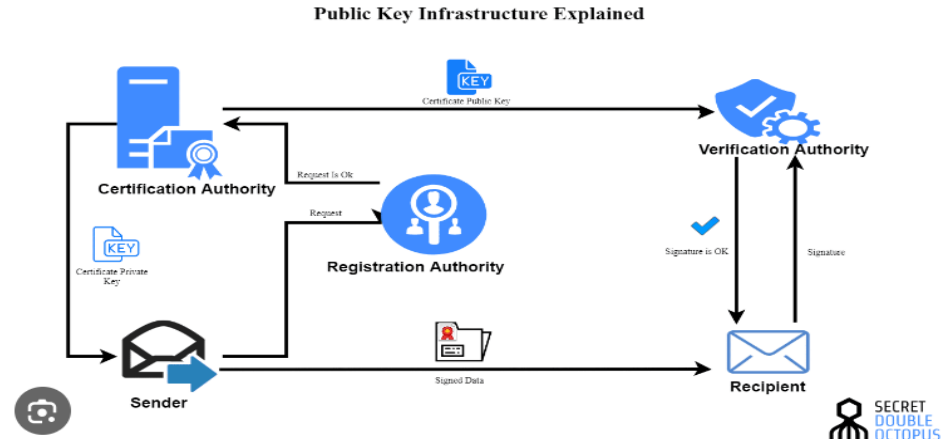
PKI / Data Encryption Standard

- A PKI stands for Public key Infrastructure in cryptography.
- The sole purpose of PKI is to facilitate the best secure electronic transfer of information for digital activities.
- PKI provides assurance of public key. It provides the identification of public keys and their distribution.



PKI / Data Encryption Standard

- An anatomy of PKI comprises of the following components.
- Public Key Certificate, commonly referred to as ‘digital certificate’.
- Private Key tokens.
- Certification Authority.
- Registration Authority.
- Certificate Management System.



PKI / Data Encryption Standard

- **DES (Data Encryption Standard)** is a encryption algorithm which uses
- symmetric keys for cipher encryption.
- It uses 56 bits (+8 parity bits) in 16 rounds having a block size of 64 bits.
- It has been designed by IBM team
- It was first published in 1975
- TRIPLE DES is a symmetric key-block cipher which applies DES cipher in triplicate. It encrypts with first key (K1), decrypts with second key (K2) then decrypts with third key (K3). This is also a two-key variant where K1 and K3 are the same keys.

PKI / Data Encryption Standard

- The main disadvantage of Public Key Infrastructure is that one of the keys i.e., public key is in a public domain and is therefore, likely to be misused.
- It is rarely seen that cryptographic schemes are compromised due to weakness in their design but very often it is compromised due to the poor key management.
- The same can be achieved by keeping private keys secret throughout.

Important Topics

- RSA and DES.
- Asymmetric or Public Key System.
- Digital certificates by Certified Authorities (CAs).
- Five common CAs in India.

Summary

- In this session, we discussed about two algorithms One is RSA and other one is DES.
- RSA is an asymmetric cryptography and DES is symmetric one.
- A system where one secret key shared is called Symmetric or Secret Key Cryptography.
- A cryptosystem where two different keys are used for encryption and decryption is called Asymmetric or Public Key System.
- Digital certificates are provided by Certified Authorities (CAs) to authenticate that a particular site is globally secured.
- There are five common CAs in India. They are Safescrypt Ltd, TCS, IDRBT, MTNL and NIC.

*Thank
You*



Security and Cyber Laws

Block-1 Unit-3

Data Security and Management

Topics to be Covered

Introduction

Database
security and
Data
Management

Security
Requirements
(CIA)

Security
Threats and
Attacks

Computer,
Mobile and
Internet

Topics to be covered

Security Measures and Solutions

Security Policy

Security Management

Security Audit

Security and Usability Summary

Important Topics

Summary

Introduction

- Data security management is a way to maintain the integrity of data and to make sure that the data is not accessible by unauthorized parties or susceptible person to corrupt it.
- Data security is put in place to ensure privacy in addition or protecting this data.
- Data security management systems focus on protecting sensitive data, like personal information or business-critical intellectual property.
- For example, data security management can involve creating information security policies, identifying security risks, and spotting and assessing security threats to IT systems.

Data Security and Data Management

- Theft and fraud
- Loss of availability of data
- Loss of confidentiality
- Loss of data privacy
- Loss of data integrity

Database security is necessary in the following situations:



Source : Shutterstock.com

Components to Data Management

Data management is the responsible stewardship of data throughout its lifecycle.

There are five components to data management:

- Acquisition
- Utilization
- Maintenance
- Access

Effective data management requires appropriate acquisition, utilisation, maintenance, access, and protection of data.

Data management depends on information confidentiality and criticality.

Security Requirements (CIA)

- Data is being used by a vast majority of individuals, entities, businesses and organizations.
- Data security is necessary and important in today's world for all devices or processes which deal with collection, management and storage of data as data breaches may occur anytime and can not only lead to litigation but also damage to the brand and reputation.
- The core elements of **data security** are **confidentiality, integrity** and **availability**.

Data Management

- The main aim of data management helps people and organizations for data to be used within the boundaries of policies and regulations.
- Data management can be achieved by the practice of collection, keeping and usage of data in a secure, efficient and cost-efficient manner.
- The management of data is done through various platforms and include databases, data analysis and more such tools like Microsoft SQL server, Google cloud, Amazon web services, etc.

Security Requirements (CIA)

The three governing principles are as follows:

- **Confidentiality**
 - Confidentiality is the ability to hide information from those people unauthorized to view it. It is the most obvious aspect of the CIA triad. Cryptography and Encryption methods are an example of an attempt to ensure confidentiality of data transferred from one computer to another.
- **Integrity**
 - The ability to ensure that data is an accurate and unchanged representation of the original secure information. One type of security attack is to intercept some important data and make changes to it before sending it on to the intended receiver.

Security Requirements (CIA)

- **Availability**

- It means information should be consistently and readily accessible for authorized parties.
- This involves properly managing hardware and technical infrastructure and systems that hold and display the information.



Source : Pan more Institute

Practices for Implementation of CIA Triad

- Some practices for implementation of CIA Triad of confidentiality, integrity and availability are as follows:
- **Putting confidentiality into practice**
 - Categorization of data and assets being handled by individuals in an
 - organization based on their privacy requirements.
 - Requirement of all data encryption and two-factor authentication to be basic security hygiene as a fundamental practice in all organizations dealing with sensitive information.
 - Ensure that access control lists and file permissions are monitored and updated regularly by professionals from the IT department in an organization.

Practices for Implementation of CIA Triad

- **Scoping integrity**

- Review all data processing, transfer and storage mechanisms and run diagnostic tests to ensure there is no unauthorized access.
- Understand organization's compliance and regulatory requirements by keeping in check with the rules and regulations updated.

- **Ensuring availability**

- Build preventive measures into system designs, make security audits routine,
- Auto update or stay alert of system, network and application updates.
- Utilize detection tools such as network/server monitoring software and anti-virus solutions and regular check-up through timely runs.

Security Threats and Attacks

- Cyber security threat refers to any possible malicious attack that seeks to unlawfully access data, disrupt digital operations or damage information.
- The following cyber security threats are constantly growing and creating issues related to data privacy:
- **Phishing**
 - Phishing attacks use fake communication, such as an email, to trick the receiver into opening it and carrying out the instructions inside, such as providing a credit card number.
 - "The goal is to steal sensitive data like credit card and login information or to install malware on the victim's machine," Cisco reports.

Security Threats and Attacks

- **Cyber-physical attacks**
 - The technology that has enabled to modernize and computerize critical infrastructure also brings risk.
 - There is an ongoing threat of hacks targeting electrical grids, transportation systems, etc., which represent a major vulnerability.
- **Malware**
 - Malware is malicious software such as spyware, ransom ware, viruses and worms.
 - Malware is activated when a user clicks on a malicious link or attachment, which leads to installing dangerous software.

Security Threats and Attacks

- **Man in the middle attack**
 - Man in the Middle (MitM) attack occurs when cyber criminals place themselves between a two-party communication.
 - Once the attacker interprets the communication, they may filter and
 - steal sensitive data and return different responses to the user.
- **State-sponsored attacks**
 - Hackers look to make profit through stealing individual and corporate
 - data.
 - Cyber crime today is a major threat not only to the private sector and
 - individuals but also towards the governments and nations as a whole.

Computer Mobile and Internet

- The computer is the foundation of the entire virtual world and is now extensively used both personally and professionally in all walks of life
- The mobile from its name itself denotes communication on the move. This has actually made many conventional systems of interaction obsolete.
- The Internet is an international network of computer systems that has evolved over the last decade. Currently, the Internet interconnects several thousand individual networks that connect over a million computers.

Computer Mobile and Internet

- Computers today have reduced complicated jobs into much simpler tasks. For example, one can write a letter in a word document, edit it, spell check, print copied and also send it to someone across the world in a mere matter of seconds.
- A computer simply is an information processor in a way that it takes whatever raw information or data which is fed by a human and stores that information, then proceeds to decrypt the information entered and consequently provide the result in the form of an output.
- The work of a computer is nothing without a computer program. We can see various computer programs on a computer we rely on like Microsoft Word, Excel, etc. used for carrying out day to day activities at all spheres of life.

Mobile

- A mobile device is a general term used for a handheld computer or a smartphone.
- The mobile devices invented not only has functions of making calls, receiving calls, sending and receiving text messages, but all contains functions of obtaining emails and carries out functions of a computer system at a smaller level.
- A mobile as per defined by digital technology refers to a cell phone usually one with computing ability, or a portable, wireless computing device used while held in the hand, as in mobile tablet, mobile, mobile app, etc.

Characteristics of Mobile

Characteristics of a mobile device

- Wi-Fi or cellular access to the Internet
- A battery that powers the device for several hours
- A physical or onscreen keyboard for entering information
- Touch-screen interface
- Ability to download data from the Internet



Characteristics of Mobile

Different meanings of mobile

In different contexts, mobiles are also defined as “mobile development”, “mobile-friendly” etc.

The term “mobile development” usually refers to creating apps for smartphones, but does not include laptops.

“Mobile friendly” on the other hand refers to websites that are easy to use by any user owning a smart phone.

Internet

- A network is a group of connected computers that are able to send data to each other.
- The Internet is a vast, sprawling collection of networks that connect to each other.
- Computers connect to each other and to the Internet via wires, cables, radio waves, and other types of networking infrastructure.
- These multiple interconnected networks form the Internet.

How does a User Access the Internet?

- A single device that is assigned with an address when it connects to the Internet known as the Internet protocol (IP) address and this address helps in differentiating between devices in the network from all other devices.
- Almost every connection to be made with the Internet requires a device which includes an address for sending/receiving messages in the form of emails.
- Mobile phones too, operate within a network based on services that are provided by service providers. They convert our voice into electronic signals which are then transmitted through radio waves.
- The use of Wi-Fi has grown two-fold due to connection to the Internet wirelessly. The concept of free Wi-Fi is now commonly available in public places such as airports, cafes, etc.

Limitations

Limitations are as below:

- Speed
- Accessibility
- Incompatibility
- Leakage of data
- Use of unsecured Wi-Fi
- SMishing



Security Measures and Solutions

- **Access security-**

- Access control is a method of restricting access to sensitive data.
- Only those that have had their identity verified can access company data through an access control gateway.

- **Data encryption-**

- Data encryption is a security method where information is encoded and can only be accessed or decrypted by a user with the correct encryption key.
- Encrypted data, also known as cipher text, appears scrambled or unreadable to a person or entity accessing without permission.

Security Measures and Solutions

- **Email security**

- It is a form of procedure to protect an email account and the contents on an email account from unauthorized access.
- Therefore, measures like strong email passwords, end-to-end encryption of emails or messages that are sent from one person to another result in prevention of misuse of data, as emails are a popular forum for hackers to spread malware, spam and phishing attacks.

Security Measures and Solutions

- **Risk-assessment analysis**

- The main of conducting a risk assessment is to identify the risks pertaining to information stored in an organizations system.
- By conducting risk assessment analysis, an organization can understand and assess internal and external risks to their security, confidentiality and personal information stored in various storage media like laptops and portable devices.

Security Measures and Solutions

- **Third party issues**
 - Website's play a major role while showcasing an organization's success.
 - Third party tools to make their websites' more interactive and user-friendly and offer smooth connectivity for user interaction.
 - These third-party tools help in generating revenue for an organization's website.
 - To ensure that all reasonable steps have been taken prior to giving access to third party service providers and apply the stringiest security measures.

Security Measures and Solutions

- **Strong firewall**

- A firewall enables to protect a system from internet traffic and services it is exposed to.
- These services are accessed by everyone who uses an internet.
- Antiviruses are for files and firewalls are needed to protect from unauthorized access or usage of network.
- A firewall simply helps to control Internet traffic that is generated by using a network for work.

Security Measures and Solutions

- **Antivirus protection-**

- An antivirus protection can be gained in the form of antivirus software.
- This software is a program designed to avoid, detect and deal with
- cyber security threats that an organization may face.
- The process of an antivirus is to run background scans on a system to detect and restrict unauthorized access in the forms of malware and to protect a system.
- These solutions are extremely important for data security and must be installed on computer systems.

Security Measures and Solutions

- **Back-up regularly**

- A data security is meant for protecting information stored on a system from unauthorized access, destruction of such information and includes network security.
- The securing of such data helps in preventing accidental modification to data, theft of data, breach of confidentiality agreements and avoid release of data prior to its verification and authentication.

Security Policy

- Security policies are a formal set of rules which is issued by an organization to ensure that the user who are authorized to access company technology and information assets comply with rules and guidelines related to the security of information.
- It is a written document in the organization which is responsible for how to protect the organizations from threats and how to handles them when they will occur.
- The purpose of the policy is to create a safe and resilient cyber space for individuals, organizations and the government.

Strategies Adopted by the Security Policy

Some of the strategies adopted by the policy include

- Creating an assertion structure;
- Encouraging open standards;
- Strengthening the administrative structure;
- Securing e-administration.



Security Management

- Security management means minimizing the interruption of business activities and reducing the vulnerability to various attacks.
- Security is not just restricted to computer systems; it applies to all perspectives of securing data or information, in whatever structure.
- There are six principles of security management:-
 - Availability
 - Integrity of data or systems
 - Confidentiality of data or systems
 - Accountability
 - Assurance
 - Privacy

Security Audit

- An IT security audit is an overall assessment of the organization's.
- A thorough audit assesses the security of the system's physical configuration and environment, software, information handling processes, and user practices.
- Security audits give a reasonable and computable way to scrutinize how secure a site really is.
- This assessment is designed to:
 - Create a security benchmark for your organization;
 - Identify the qualities and shortcomings of current security rehearses;
 - Prioritize the exposures that present the most serious hazard;

Security and Usability

- The term usability can be taken in narrow terms of quality of a system's interface, but the concept applies more broadly to how a system supports the requirements of the user.
- Usability though dealing with quality of a system's interface also includes the term “user experience”.
- Definition of usability is: **“the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use.”**

Usability

- Usability deals with the following outcomes:
- A website or product should be easy for a user, so that they can navigate such websites.
- A user can achieve their objectives through using a particular website by way of easy and detailed navigation.
- Usability though dealing with quality of a system's interface also includes the term “user experience”.
- This refers to the ease with which a user can access or use a product or a website.

Important Topics

Database security and Data Management.

Security Threats and Attacks online and offline.

Security Measures and Solutions.

Summary

- In this Session, we learnt about the protection of data from unauthorized access, use, change, disclosure and destruction by using methods to ensure network security, physical security and file security based on a collection of standards and technologies that protect data from intentional or accidental destruction, modification or disclosure is known as data security.
- Data security can be applied through various techniques and technologies including administrative controls, organizational standards, etc.

Summary

- Database security is necessary for the following situations:
- Theft and fraud
- Loss of confidentiality or secrecy
- Loss of data privacy
- Loss of data integrity
- Loss of availability of data
- Data management is the practice of collecting, keeping and using data securely, efficiently, and cost-effectively.
- The core elements of data security are confidentiality, integrity and availability. This is a security model and guide for organizations to keep their sensitive data protected from unauthorized access and data exfiltration.

*Thank
You*



Security and Cyber Laws

Block-1 Unit-4

Regulation of Cyberspace

Topics to be Covered

Introduction

Desirability of Regulation of Cyberspace

Need for Regulation of Cyberspace

How cyberspace can be regulated

Legal and Self Regulatory Framework

Filtering Devices and rating systems.

Government policies and Laws regarding regulation of Internet Content

UNCITRAL MODEL LAW

Regulation of cyberspace content : Global Scenario

United States

Topics to be Covered

Communication Decency Act (CDA)

COPA

CIPA

Other Related Legislations

European Union

United Kingdom

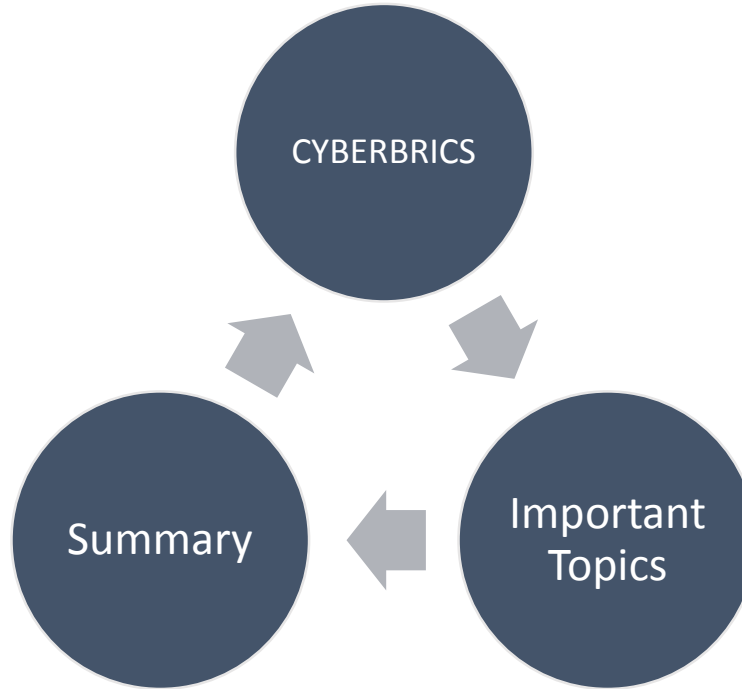
Regulation of Cyberspace content in India

International initiatives for regulation of Cyberspace

Organization for Economic Cooperation and Development (OECD)

UNESCO

Topics to be Covered



Introduction

- Cyberspace refers to the virtual computer world, and more specifically, is an electronic medium used to form a global computer network to facilitate online communication.
- It is a large computer network made up of many worldwide computer networks.
- Cyberspace's core feature is an interactive and virtual environment for a broad range of participants.



Desirability of Regulation of Cyberspace

- Cyberspace spans worldwide, but it has no formal framework. It has no definite metes and bounds except the capacity of the hardware used for access.
- Regulation in cyberspace is an emerging challenge. Anonymity encourages and enhances the exercise of freedom. A child too shy to express himself in the physical space can feign to be somebody else in virtual space, and express himself freely.
- The inventions of new technologies, the media has enhanced the possibility of invasion of the privacy of individual and bringing it into the public domain.
- The internet also provides speed and ease of transmission of both voice and data. The facility in communication greatly enhances global trade.

Desirability of Regulation of Cyberspace

- Encryption is the process of converting a message or document into a form which hides the content of the communication from the eyes of an eavesdropping third party and needs to be decrypted if its content is to be read.
- Internet has emerged as the ‘media of the people’ as the internet spreads fast there were changes in the press environment that was centered on mass media.
- Viruses, rumor-mongering, hate-mail and mail box bombardment are all describable phenomena and because of the fear of retribution all are more likely to use fake identity or may be anonymous mailers rather than a readily identifiable person. Therefore, Cyberspace needs to be regulated to curb this phenomenon.

How Cyberspace can be Regulated

- The hardware and software of Cyberspace that define the system can be a form of regulation.
- It is a set of rules implemented or codified in the software by the code writers, requiring the constant certification of identity.
- Regulation of the internet can increasingly rely on a three-fold strategy:
 - Privatization:
 - Propertization:
 - Technological Controls the system is to be designed so as to hardware in desired regulatory features.
- Blocking software or Internet contents grading system are other forms of regulations based on technology.

Legal and Self-Regulatory Framework

- Self-regulation emerges as very important form for regulating different aspects of the electronic domain.
- In this context of regulation of cyberspace, it can be said that the
- Internets' design precludes central control which may be regulated by government to make the information economy safe, secure, certain and open.
- It has been argued that the selective government regulation of cyberspace is warranted to protect and promote liberal democratic ideas.

Filtering Devices and Rating Systems

- ‘Filters’ are software tools used to block access to unwanted material. For example, E-mail filters automatically deletes the bulk of unread e-mail messages commonly known as ‘spam’ and can also be customized to delete incoming messages from particular sources.
- Site blocking filters also may be used a protocol ‘PICS’ developed by the World Wide Web Consortium (‘W3C’) to develop common protocols for the World Wide Web’s evolution and ensure its interoperability.

Filtering Devices and Rating Systems

- Organizations in several countries have established labeling schemes, which conform to the PICS standards, designed for use by parents and schools.
- In 1997 W3C created the ‘Metadata Activity’, which includes the Resource Description Framework (RDF) Working Group.
- RDF is a protocol for description of Internet content based on a set of 105 ‘categories’ of information, known as the ‘Dublin Core’, which is used to Filter out obscene content.

Government Policies And Laws Regarding Regulation Of Internet Content

- Government policies can be classified into the following four categories:
- The policy to encourage self-regulation.
- Criminal law penalties applicable to content providers who make content “unsuitable for minors” available online.
- The government has also mandated blocking of access to content deemed unsuitable for adults; for example: Australia, China, Saudi Arabia, Singapore etc.
- A number of countries have either prohibited general public access to the internet or require internet users to be a registered / licensed by a government authority before permitting them restricted access.

UNCITRAL Model Law, 1996

- The UNCITRAL Model Law on Electronic Commerce was adopted by the
- United Nations Commission on International Trade Law (UNCITRAL) in 1996.
- The specific goal is to eliminate all kinds of legal and other technical kinds of hindrances.
- And to boost legal certainty for the cyber domain has postulated a series of internationally acceptable rules though it does not regulate every single phase of transactions done electronically.
- It has embraced the following doctrines:
 - Non discrimination principle
 - Technological neutrality principle
 - The last is the equivalence functional principle

UNCITRAL Model Law, 1996

- All the states have tries to incorporate the UNCITRAL rules so as to bring uniformity in the laws.
- This Model Law has been divided into two parts:
 - Part 1 deals with general provisions regarding e-commerce
 - Part 2 deals with specific provisions for e-commerce in certain areas.



United Nations
UNCITRAL

Source : <https://blog.ipleaders.in/>

Regulation of Cyberspace Content: Global Scenario

United States

- Cyber regulation in the United States is composed of assorted laws, executive orders, court rulings, government programs, technological standards and national security directives, which were added during the last thirty years.
- The starting point for understanding the accumulation of federal cyber regulations over time is the Comprehensive Crime Control Act of 1984.

Regulation of Cyberspace Content: Global Scenario

Communication Decency Act 1996 (CDA)

- The CDA created a criminal cause of action against those who knowingly transmit “obscene” or “indecent” messages, as determined by
- local community standards, to a recipient under the age of 18 years.
- The provisions also prohibited a person from knowingly permitting a telecommunications facility under that person’s control to be used to commit these offences.

CIPA

- In 2000, Children Internet Protection Act (CIPA) was passed. This Act requires the schools and libraries to install filters on computers used by minors and adults.

Regulation of Cyberspace Content: Global Scenario

Other Related Legislation

- Uniform Electronic Transaction Act, 1999 (UETA)
- Uniform Computer Information Transaction Act, 2000 (UCITA)
- Health Insurance Portability and Accountability Act of 1996 (HIPAA)
- Fair Credit Reporting Act, 1970



Source : <https://www.e-ir.info/>

European Union

- The European Union is a framework of democratic countries in Europe, which originated around creating a common European market.
- Three main action lines;
 - Creating a safer environment through promotion of hotlines, encouragement of self-regulation and codes of conduct,
 - Developing filtering and rating systems, facilitation of international agreement on rating systems.

European Union

- **Awareness:** Making parents, teachers and children aware of the potential of the Internet and its drawbacks, overall co-ordination and exchange of experience.
- The word ‘cyber security’, from the viewpoint of European Union, involves a blend of cyber resilience, cybercrime, cyber defense, cyber security and global cyberspace concerns.



Source: freepik.com

European Union

- In September 1996, UK Government issued R3 Safety-Net action plan developed by UK ISP trade associations.
- The UK's cyber security mission is led by the National Cyber Security Centre (NCSC), which is a part of GCHQ.
- The NCSC helps protect the UK's critical services from cyber attacks, manages major incidents, and improves the underlying security of the UK Internet through technological improvement and advice to citizens and organizations.
- It supports the most critical organizations in the UK, the wider public sector, industry and small and medium enterprises (SMEs).

Regulation of cyberspace content in India

- In India, Information Technology Act, 2000 is the legislation which covers the domain of cyber law.
- The main objective of the Act is to provide legal recognition for transactions carried out by means of electronic data interchange and electronic communication, commonly referred to as ecommerce, which involve the use of alternatives to paper-based methods of communication and storage of information to facilitate electronic filing of documents with the Government agencies.

Regulation of Cyberspace Content in India

- Electronic Signatures
- Electronic Governance
- Regulation of Certifying Authorities
- Digital Signature Certificate
- Penalties and Adjudication
- The Appellate Tribunal
- Offences



International Initiatives for Regulation of Cyberspace

- In July 1997, the German Government hosted an
- International Conference in Bonn on the topic ‘Global Information Networks’, in cooperation with the European Commission that resulted in the adoption of the ‘Bonn Declaration’ of the Ministers as well as declarations by industry and user participants. The Bonn Declaration pointed in the direction of:
 - using current national legal frameworks for the enforcement of criminal
 - law provisions where appropriate in respect of on-line crime;
 - development by industry of common principles for schemes of self-regulation regarding content of on-line services; and
 - establishment of national hotlines for complaints regarding on-line content and for some appropriate interconnection and interaction between national hotlines.

Organization for Economic Cooperation and Development (OECD)

- The OECD an international organization working in the area of data privacy and information security, established an ad hoc process of meetings (the first was on 1-2 July 1997 and second on 22 October 1997).
- The meeting acknowledged the primary role of the private sector in regulating the Internet.
- On 19 April 2006, OECD task force on spam has recommended that Governments and industry should step up their coordination to combat the global problem of spam.
- Co-ordination and co-operation between public and private sectors are critical, the report notes.

UNESCO

- The United Nations Educational, Scientific and Cultural Organization (UNESCO) was founded on 16 November 1945.
- At the 29th UNESCO General Conference held in Paris from 21 October to 12 November 1997.
- Today, UNESCO functions as a laboratory of ideas and a standard-setter to forge universal agreements on emerging ethical issues:
 - The organization also serves as a clearing house for the dissemination and sharing of information and knowledge
 - It helps Member States to build their human and institutional capacities in diverse fields.

Cyberbrics

- This project has a threefold purpose.
- Firstly, to plot the prevailing protocols; to recognize finest procedures or ways and build policy propositions in the arena of cyber security governance, Internet access policy and tactics for the digitization of public supervision in the BRICS (Brazil, Russia, India, China and South Africa)
- The focus areas are: data protection, consumer protection, cybercrime, protection of public order and lastly cyber defense.

Important Topics

Internet for
commercial
transactions and
consumer services.

Misuse of internet

Legal policies.

Summary

- In this session, we have learnt about use of internet for various types of commercial transactions and consumer services.
- The necessity arises due to the expansion of economic and creative interaction which in term led to disputes involving acquisition, use, possession, processing and communication of information.
- The use of internet for obscene, indecent and pornographic content, rumor mongering, viruses, cyber crime, possibility of invasion of privacy of individuals, all this emphasized the need for cyberspace regulation.
- Legal policies in various countries like USA, UK, European Union, and New Zealand show that in the context of regulation of Cyberspace more emphasis is on self regulation through use of filtering/blocking technologies.

*Thank
You*

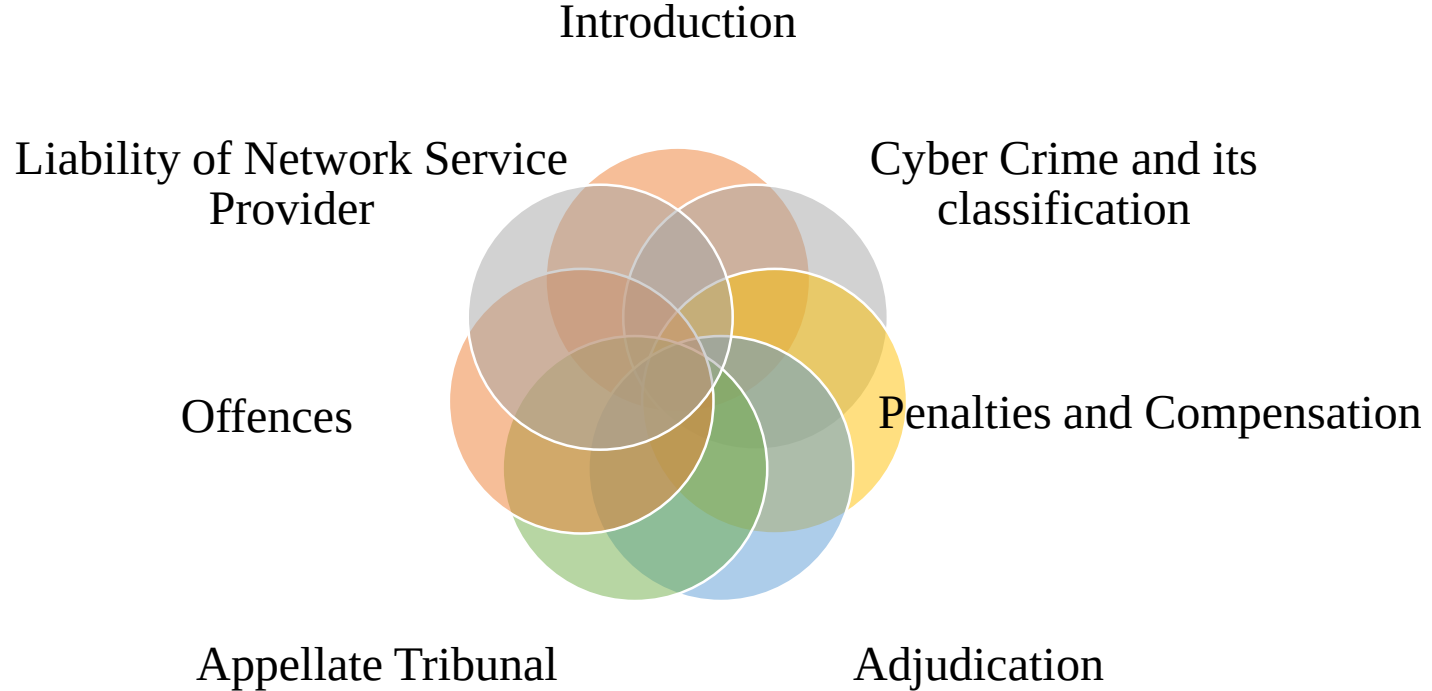


Security and Cyber Laws

Block - 2 Unit - 5

Cyber Crimes

Topics to be Covered



Introduction

- The purpose of the IT Act is to provide legal recognition for transactions carried out by means of electronic data interchange and other means of electronic communication, commonly referred to as electronic commerce, which involve the use of alternatives to paper - based methods of communication and storage of information.
- It also aims at facilitating electronic filing of documents with the Government agencies.



Source :<https://www.alamy.com/>

Introduction

- Need was felt for uniformity of the law applicable to alternatives to paper-based methods of communication and storage of information; The act has undergone several amendments since its enactment in the year 2000.
- The most significant amendment is of 2008 which tried to address a number of issues keeping in view the technological development which facilitated the commission of certain offences which had a greater impact on the society.
- In this session the wrongful acts covered in the act will be discussed. Some wrongful acts are civil wrong for which the aggrieved party is entitled to penalty or compensation. Some acts are treated as crime for which punishment is provided.

Cyber Crime and Its Classification

- Cybercrime is defined as crimes committed on the internet using the computer as a tool to target the victim for the execution of the desired crime.
- It is difficult to determine that where the particular cyber crime took place because it can harm its victim even sitting at a far distance.
- As stated above from the year 1997 to 2008 tremendous changes took place which helps the judicial system to determine the specific kind of cyber crime.
- All cybercrimes involved both the computer and the person behind it as victims, it just depends on which of the two is the main target.

Cyber Crime and Its Classification

- Example 1: Hacking involves attacking the computer's information and other resources.
- Example 2: Stalking involves attacking the personal space of an individual.



Cyber Crime and Its Classification

- The cyber crimes are broadly classified into different groups:
- Crime against the individuals – Harassment, cyber-stalking, deformation, indecent exposure, cheating, email spoofing, fraud, etc.
- Crime against property – Transmitting virus, net-trespass, unauthorized control over computer system, internet thefts, infringement of intellectual property, etc.
- Crime against organization – Cyber terrorism within government organization, possession of unauthorized information, distribution of pirate software, etc.

Cyber Crime and Its Classification

- Crime against society – Child pornography, financial crimes, sale of unlawful articles, trafficking, forgery of records, gambling, etc.
- Cyber crimes are quite different from traditional crimes as they are often harder to detect, investigate and prosecute and because of that cyber crimes cause greater damage to society than traditional crimes.
- Cyber crime also includes traditional crimes conducted through the internet or any other computer technology. For example; defamation, forgery, identity theft, terrorism, cyber-stalking, hacking, software piracy, web jacking and bullying are considered to be cyber crimes when traditional crimes are committed through the use of a computer and the internet.

Penalty and Compensation

- Section 43 to 45 of Information Technology Act, 2000 provides for the instances where the wrong doer is liable to pay damages by way of compensation to the effected party.
- Section 66 of Information Technology Act, 2000 however provides that if any person, dishonestly or fraudulently, does any act referred to in section 43, he or shall be punishable with imprisonment for a term which may extend to three years or with fine which may extend to five lakh rupees or with both.



Penalty and Compensation

- Section 43 of Information Technology Act, 2000 provides that any person who without permission of the owner or any other person who is in charge of a computer, computer system or computer network commits the following acts shall be liable to pay damages
- Accesses or secures access.
- Downloads, copies or extracts any data, computer data base or information.
- Introduces or causes to be introduced any computer contaminant or computer virus.

Penalty and Compensation

- Explanation to this section provides:
- “Computer contaminant means any set of computer instructions that are designed:
 - To modify, destroy, record, transmit data or program residing within a computer, computer system or computer network;
 - By any means to usurp the normal operation of the computer, computer system, or computer network;



www.istockphoto.com/photos/penalty-fee

Penalty and Compensation

- Computer virus means any computer instruction, information, data or program that destroys, damages, degrades or adversely affects the performance of a computer resource or attaches itself to another computer resource and operates when a program, data or instruction is executed or some other event takes place in that computer resource;”
- Damages or causes to damaged data, computer data base or any other programs; Explanation.
 - This section provides, “ damage means to destroy, alter, delete, add, modify or rearrange any computer resource by any means.”

Penalty and Compensation

- Disrupts or causes disruption
- Denies or causes the denial of access to any person authorized to access by any means;
- Provides any assistance to any person to facilitate access in contravention of the provisions of this Act, rules or regulations made there under;
- Charges the services availed of by a person to the account of another person by tampering with or manipulating;
- Destroys, deletes or alters any information residing in a computer resource or diminishes its value or utility or affects it injuriously by any means;

Penalty and Compensation

- Steal, conceal, destroys or alters or causes any person to steal, conceal, destroy or alter any computer source code used for a computer resource with an intention to cause damage;
- [computer source code|| means the listing of program, computer commands, design and layout and program analysis of computer resource in any form.]
- Section 43A of Information Technology Act, 2000 provides for the liability for Compensation of a body corporate for failure to protect data.
- Section 44 of Information Technology Act, 2000 provides for penalty for failure to furnish information, return, etc.

Penalty and Compensation

- “If any person who is required under this Act or any rules or regulations made there under to—
- Furnish any document, return or report to the Controller or the Certifying Authority fails to furnish the same, he shall be liable to a penalty not exceeding one lakh and fifty thousand rupees for each such failure;
- File any return or furnish any information, books or other documents within the time specified therefore in the regulations fails to file return or furnish the same within the time specified therefore in the regulations, he shall be liable to a penalty not exceeding five thousand rupees for every day during which such failure continues;

Penalty and Compensation

- Maintain books of account or records, fails to maintain the same, he shall be liable to a penalty not exceeding ten thousand rupees for every day during which the failure continues.”
- Section 45 of Information Technology Act, 2000 provides for the residuary penalty. Contravention of any rules or regulations made under this Act, for the contravention of which no penalty has been separately provided. The maximum penalty in such cases is 25000 rupees.
- **Adjudication**
 - Section 46 provides for the adjudication of disputes for awarding compensation.

Penalty and Compensation

- This officer shall exercise jurisdiction to adjudicate matters in which the claim for injury or damage does not exceed rupees five crore.
- A reasonable opportunity for making representation in the matter and on such inquiry must be given.
- Where it accedes 05 (five) crore, The jurisdiction shall vest with the competent civil court:
 - All proceedings before adjudicating officer shall be deemed to be judicial proceedings within the meaning of sections 193 and 228 of the Indian Penal Code, 1860;

Penalty and Compensation

- It shall be deemed to be a civil court for the purposes of sections 345 and 346 of the Code of Criminal Procedure, 1973;
- It shall be deemed to be a civil court for purposes of Order XXI of the Civil Procedure Code, 1908.
- No person shall be eligible to be appointed as an adjudicating officer if he does not possess such experience in the field of Information Technology and legal or judicial experience which is explicitly prescribed by the Central Government.

Penalty and Compensation

Appellate Tribunal

- An appeal tribunal is a special court or committee that is formed to reconsider a decision made by another court or committee.
- Section 48 provides that from coming into force of the Finance Act, 2017 Telecom Disputes Settlement and Appellate Tribunal established under section 14 of the Telecom Regulatory Authority of India Act, 1997), shall be the Appellate Tribunal for the purposes of this Act and the said Appellate Tribunal shall exercise the jurisdiction, powers and authority conferred on it by or under this Act.

Penalty and Compensation

- Section 57 provides the procedure for appeal. Any person aggrieved by an order made by controller or an adjudicating officer under this Act may prefer an appeal to a Appellate Tribunal having jurisdiction in the matter.
- Every appeal shall be filed within a period of forty-five days from the date on which a copy of the order made by the Controller or the adjudicating officer is received by the person aggrieved and it shall be in such form and be accompanied by such fee as may be prescribed:
- Provided that Appellate Tribunal may entertain an appeal after the expiry of the said period of forty-five days if it is satisfied that there was sufficient cause for not filing it within that period.

Penalty and Compensation

- Section 58 provides that Appellate Tribunal shall not be bound by the procedure laid down by the Code of Civil Procedure, 1908.
- But It shall be guided by the principles of natural justice and, subject to the other provisions of this Act and of any rules, it shall have, for the purposes of discharging its functions under this Act.
- The same powers as are vested in a civil court under the Code of Civil Procedure, 1908 (5 of 1908), while trying a suit, in respect of the following matters, namely:—
- (a) Summoning and enforcing the attendance of any person and examining him on oath;

Penalty and Compensation

- Section 62 provides for the appeal to high court. Any person aggrieved by any decision or order of the Appellate Tribunal may file an appeal to the High Court within sixty days from the date of communication of the decision or order on any question of fact or law arising out of such order:
- Provided that the High Court may, if it is satisfied that the appellant was prevented by sufficient cause from filing the appeal within the said period, allow it to be filed within a further period not exceeding sixty days.

Offences

Following acts are considered as offences under the act:

- Section 65 provides for tampering, concealing, destroying , or altering any computer source document intentionally. Penalty is up to Rs.2,00,000/-, or Imprisonment up to 03 (three) years, or both.
- Section 66 provides for dishonestly, or fraudulently doing any act as referred in Section 43. Penalty is up to Rs.5,00,000/-, or Imprisonment up to 03 (three) years, or both.
- Section 66A has been struck down by Supreme Court's Order dated 24th March, 2015 in the landmark precedent of "*Shreya Singhal vs. Union of India*", AIR 2015 SC.1523.25. The court found it as violation of the Freedom of Speech and Expression guaranteed under the Constitution of India.

Offences

- Section 66B provides for dishonestly, or fraudulently receiving or retaining any stolen computer resource or communication device. Penalty is up to Rs.1,00,000/-, or Imprisonment up to 03 (three) years, or both.
- Section 411 of the IPC, 1860 provides punishment for dishonestly receiving stolen property and is worded in a manner that is almost identical to section 66B of the IT Act. The punishment under section 411 of the IPC is imprisonment of either description for a term of up to 3 years,
- Section 66C provides for dishonestly, or fraudulently making use of Electronic Signature, Password or any other Unique Identification Feature of any other person. Penalty is up to Rs.1,00,000/-, or Imprisonment up to 03 (three) years, or both.

Offences

- Section 66D provides for dishonestly, or fraudulently by means of any communication device or computer resource cheating by personating. Penalty up to Rs.1,00,000/-, or Imprisonment up to 03 (three) years, or both.
- Section 419 of the IPC, 1860 provides punishment for 'cheating by personating' and provides that any person who cheats by personating shall be punished with imprisonment of either description for a term which may extend to 03 (three) years or with a fine or with both.

Offences

- Section 420 of the IPC, 1860 provides for any person who cheats and thereby dishonestly induces the person deceived to deliver any property to any person, or to make, alter or destroy the whole or any part of a valuable security, or anything which is signed or sealed, and which is capable of being converted into a valuable security shall be punished with imprisonment of either description for a term which may extend to 07 (seven) years, and shall also be liable to fine.
- Section 66E of Information Technology Act, 2000 provides for intentionally capturing, publishing, or transmitting image of private area of any person without consent. Penalty is up to Rs.2,00,000/-, or Imprisonment up to 03 (three) years, or both.’ etc.

Offences

Liability of Network Service Providers

- A ‘network service provider’ means any person who provides access to information service in electronic form. For example: Internet service provider, cellular mobile services, customer access services, mobile satellite services etc.
- It essentially performs two tasks-to provide access to the network and to act as intermediary between an originator and addressee with respect to any particular electronic message.

Offences

- The function of the intermediary is limited to providing access to a communication system over which information made available by third parties is transmitted or temporarily stored or hosted; or
- The intermediary does not
 - Initiate the transmission,
 - Select the receiver of the transmission, and
 - Select or modify the information contained in the transmission;
- The intermediary observes due diligence while discharging his duties under this Act and also observes such other guidelines as the Central Government may prescribe in this behalf.

Offences

Liability of Network Service Providers

- The above exemption shall not apply if–
- The intermediary has conspired or abetted or aided or induced, whether by threats or promise or otherwise in the commission of the unlawful act;
- After receiving actual knowledge, or on being notified by the appropriate government or its agency that any information, data or communication link residing in or connected to a computer resource controlled by the intermediary is being used to commit the unlawful act, the intermediary fails to expeditiously remove or disable access to that material on that resource without vitiating the evidence in any manner.

Investigation

- For the purpose of conducting cyber-crime investigation, essential special skills and technical tools are required without which the investigation is next to impossible.
- After commencement of the Information Technology Act, 2000, some provisions of Criminal Procedure Code, 1973 and the Evidence Act, 1872 have been duly amended.
- Along with these, certain new rules and regulations had been enforced by the Indian legislative system to meet the need of cyber-crime investigation.

Investigation

- Section 75 deal with the issue of jurisdiction with respect to cyber crimes. As we know, cyber crime knows no boundary. A person sitting in one country can commit offences having its consequences in another country.
- Section 75 provides that if any person have committed an offence, or contravention committed outside India, and if the act or conduct constituting the offence or contravention involves a computer, computer system or computer network located in India, then the provisions of this Act shall apply also to any offence or contravention committed outside India by any person irrespective of his nationality etc.

Cyber Forensics

- Cyber forensics also known as computer forensics which is the application of investigation and analysis techniques to gather and store evidence from a particular computing system in a way that is appropriate for presentation in a court of law.
- “Cyber forensics is the process of acquisition, authentication, analysis and documentation of evidence retrieved from the systems or online used to commit the crime.
- The systems could be from computers, networks, digital media or storage devices that could contain valuable information for the investigators to examine.

Cyber Forensics

- From online, it could be from e-commerce domains or other websites. In cyber forensics, file or data carving techniques are most commonly used to extract digital evidence from the source, hard drive or online domain.
- Computer forensics is important not just because it does recover files hidden or deleted away from storage devices and systems but it can also tell forensics experts whether are there any suspicious activities going on or had the systems been tampered with.

Cyber Forensics

- Computer forensics had helped solved the issue of recovering information from files where file system is unavailable or file system structure is corrupted.
- Files may be intentionally deleted or worse formatted to the interest of the suspect to conceal his actions.
- In today's modern era where technology plays a part in almost all the electronic devices, it is important to know when required, how a trained forensics specialist can perform up to expectation, in collecting and present his evidence findings to corresponding agencies.
- Examinations of forensics evidence are normally held in forensics laboratories or clean rooms by computer forensics investigators.

Cyber Forensics Investigation Tools



Cyber forensic techniques include:

- Cross-driven analysis that correlates data from multiple hard drives.
- Live analysis, which obtains data acquisitions before a PC is shut down.
- Deleted file recovery.
- Detecting data theft using Stochastic Forensics.
- Concealing a file, message, image, or video within another file using Steganography.

Cyber Forensics Investigation Tools

Computer forensic investigations go through five major standard digital forensic phases:

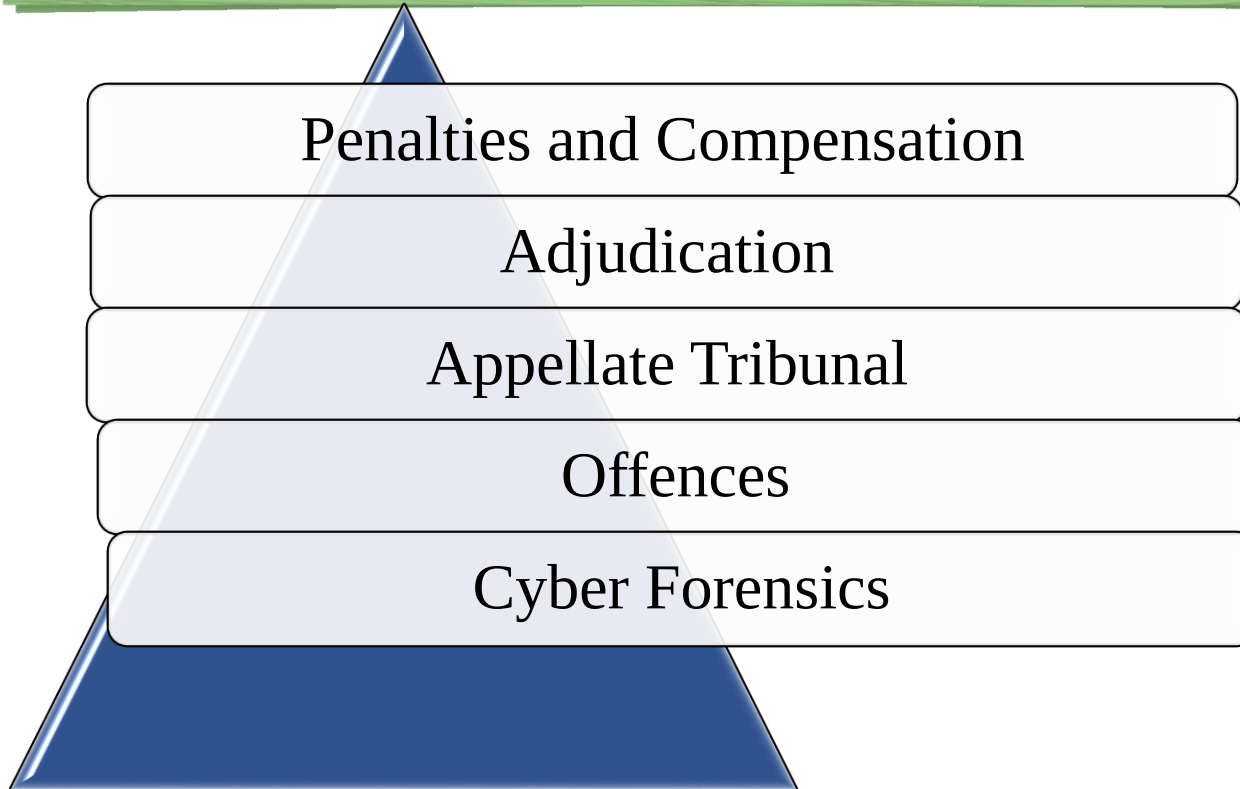
- Policy and procedure development,
- Assessment,
- Acquisition,
- Examination
- Reporting.



Cyber Forensics Investigation Tools

- Digital evidence is so fragile, it can be easily damaged, modified or destroyed purposely. That is why most of the time, original evidence are often duplicated and analysis is carried out on the duplicated copy to prevent any mishap of damaging the original copy.
- Scope of digital evidence examination can be very broad, it can be either online or offline. Examples of them are credit card transactions, Internet communications history, hard drives and other storage devices.
- Digital evidence is very critical to an investigation because the information on the evidence can tell the investigator what really happened and pieced together the whole picture.

Important Topics



Summary

- In this session, we discussed the penalties and offences provided under the Information Technology Act, 2000. Any type of unauthorized intrusion in the computer, computer system or network is prohibited and any person who does it is liable to pay compensation.
- The penalties include: introducing viruses in the computer system, unauthorized download of copyrighted material, charging services in the name of others etc.
- Offences covered in the act are mainly related to hacking, misrepresentation, identity theft, publishing obscene material, child pornography, unauthorized access to protected system etc.

*Thank
You*



Security and Cyber Laws

Block-2 Unit-6

IPR Issues in Cyber Space

Topics to be Covered

Introduction

Basic Concept: IPR

Forms of IPR

Copyright Issues in Digital Medium, Music and Goods

Patent Misuse

Linking In-lining and Framing

Trade Mark Issues

Domain Name Disputes-cyber squatting

Topics to be Covered

Search Engines and their abuse

Regulatory Framework – National and International Scenarios

Legal Protection in India

International Scenario

Important Topics

Summary

Introduction

- Management of Intellectual Property Rights in cyberspace is an important issue to combat property infringements in the virtual space ensuring security to the Intellectual Property Rights holders that they can control the use of their intellectual property and be protected from unauthorized or unlicensed use of literary work, trademarks, trade names, service marks, images, music or sound, piracy of software's.
- The Infringements in digital media may take different forms that include Copyright's violations, Deep Hyper linking, Framing, Meta-tags, spamming.

Basic Concept: IPRS

- The term Intellectual Property can be defined as intangible property which is creations of one's mind and is not merely an idea but an expression of it wiz; musical, literary and artistic works; inventions; designs; symbols, names and images.
- Novelty is considered as the main ingredient for fulfilling the condition of the intellectual property.
- The rationale behind providing Intellectual Property rights and legal protection to the creators and inventors is to give them, the due recognition for their intellectual work and also the monetary benefits for certain period of time.

Basic Concept: IPRS

- There are basically seven forms of intellectual property: copyright and related rights; trademarks, patent, industrial design, geographical indications, trade secrets and plant variety. (<https://www.wipo.int>).

I. Copyright and related rights

- A copyright is used to protect creative literary, musical, dramatic, or other artistic works like cinematographs films and sound recordings inclusive of musical compositions, audio recordings, paintings, photos, sculptures, books, articles, diagrams, movies, website content and even computer software and programs though the inventions related to software are protected under patent law.

Basic Concept: IPRS

Forms of IPR

- **Rights Granted:** provides economic and moral rights. The copyright owner has exclusive rights pertaining to reproduction and distribution of their literary and artistic work; Public performance of the work; Broadcasting of the work; communicating the work to public by wire or wireless means; Commercial rental of the work.
- **Copyright protection is available** if the work is original and exists in some tangible form based on the national laws. It does not necessarily require registration except for evidence that establishes ownership. The right is granted for specific period and may vary from country to country and from a particular class of work to another class of work.

Basic Concept: IPRS

Forms of IPR

II. Patents

- A patent is an exclusive right granted to protect an invention which is a product or a process and can also be applicable to newly engineered plant species or strain however a discovery, scientific theory or mathematical method is excluded from patentability, but its application or use can be patentable.
- **Rights Granted:** are territorial in nature and patent protection is granted for a limited period.
- **Patentable and non-patentable inventions** -An invention is patentable, if it is Novel, has Inventive step (non- obvious) and capable of industrially application.

Basic Concept: IPRS

Forms of IPR

- The methods of doing something like book keeping, trading of stocks; Diagnostic, therapeutic and surgical methods for the treatment of humans and animals; Inventions contrary to humanity, public order, morality, public health, environment and safety are not patentable. For example, process of cloning.
- There are certain products or process which are Novel and have Industrial application and are protected as **Utility Model not as patent**, for a shorter period, generally 10 years and do not require inventive step as the protection requirement like patents.

Basic Concept: IPRS

Forms of IPR

III. Trademarks

- A trademark is a distinctive sign, word, symbol or mark used in trade to distinguish the goods or services. Trademarks help consumers to identify the source of products or services. It could be name, signature, logo, brand label, phrase, slogan, letter, a numeral or any combination of them.
- **For registration of trademark**, it should be (i) distinctive in nature (distinguishable from other goods and service). Registration of trade mark is valid for specific period and needs to be renewed.

Basic Concept: IPRS

Forms of IPR

- **Registered trademark owner** has exclusive right to use and is also entitled to complete or partial assignments of rights in relation to the mark to another person, including the right to earn royalties. Trade mark owner can also permit restricted use of trademark by way of trademark licensing.
- **Industrial designs** Consists of appearance of a product/logo; the shape of an object; composition of design, pattern of cloth. The industrial design may have three-dimensional features, such as the shape or surface of an article, or two-dimensional features, such as patterns, lines or color.

Basic Concept: IPRS

Forms of IPR

- **For protection**, an industrial design must be (i) new or original and (ii) aesthetic and requires mandatory registration.

V. Geographical Indication (GI)

- It is a name or sign used on certain products which corresponds to a specific geographical location or origin (e.g. a region, or country). A Geographical Indication should have special quality or reputation.
- Geographical indications are typically used for agricultural products, food stuffs, wine and spirit drinks, handicrafts, and industrial products. Examples: Basmati rice, Swiss watches, Ethiopian coffee, Tequila for spirits produced in Mexico, Electrical appliance Made in UK.

Basic Concept: IPRS

Forms of IPR

- In order to function as a GI, a sign must identify a product as originating in a given place and the qualities, characteristics or reputation of the product should essentially be due to the place of origin.

VI. Trade Secrets

- A **trade secret** is any confidential secret information having inherent economic advantage to company and is used in business that gives a competitive edge by reason of it being secret. Examples include formulae, practice, program, process, recipes, pattern, technique, compilation, method, and device or product mechanism.

Basic Concept: IPRS

Forms of IPR

- **To qualify trade secret protection**, no registration is required. However, to protect a trade secret having commercial value the businesses must limit the number of persons who knows or access the information and get the non-disclosure agreements signed by employees.
- Trade secret remains valid as long as one does not discover it independently.

Basic Concept: IPRS

Forms of IPR

- But now a days Geographical indication is also gaining a lot of attention as it is related to specific region or any specific nature of work and by protecting those specific regional work the intellectual rights of those regional people are protected and this protection is promoting the work of all these regional people.
- The Information Technology Act, 2000 does not mention any thing about intellectual property rights. On the other hand, it can be taken into consideration that infringement of intellectual property rights is a very common practice in the cyberspace and it is very easy to practice any kind of infringement over cyberspace.

Basic Concept: IPRS

Forms of IPR

- There are some categories of intellectual property that needs law to regulate the protection of intellectual works in cyberspace. They are:
 - Intellectual works in digital form can easily be replicated.
 - Intellectual works in digital form can easily be transmitted.
 - Intellectual works in digital form can easily be modified and manipulated.
 - Intellectual works in digital form can easily be form in resemblance or equivalence can easily be created.
 - One can easily search anything in digital space and link their own work with someone else's work.

Basic Concept: IPRS

- Apart from the infringements stated before, Intellectual property infringements in cyberspace comprise of any unauthorized or unlicensed use of:
- Trademarks,
- Trade names,
- Service marks,
- Images,
- Music or sound or literary matter.



Origin, Scope, Need, Types of Intellectual
Property Rights(IPR)

Source : blog.ipleaders.in/

Copyright Issues in Digital – Medium, Music and Goods

- The global usage of Internet makes it feasible for any user to share information in cyberspace through various social media means leading to various concerns and issues related to piracy and counterfeited goods leading to huge monetary losses and as a result fake and pirated products in market.
- Gulla, R. K., 2007 has rightly pointed out that “the Internet in a way presents a troublesome situation for copyright holders as the users become mass disseminators of others copyright material and creates disequilibrium between the authors and users”.

Patent Misuse

- A patent ensures total protection of the patented invention under the legal system of a country under the legislation of which it is obtained.
- A patent represents monopoly to the patentee that they have the exclusive right to presents to the public the knowledge they have and the patented invention can't be commercially used by any one, or made, distributed or sold without the consent of patent holder.
- The object of the grant of Patent is to encourage research and development, new innovation and industrial progress. Any kind of practical application in the computer device is known to be patentable but not all software's are patentable.

Linking, In-lining And Framing

- Linking" allows a Web site user to visit another location on the Internet. By simply clicking on a "live" word or image in one Web page, the user can view another Web page elsewhere in the world, or simply elsewhere on the same server as the original page.
- This technique is what gives the Web its unique communicative power. At the same time, however, linking may undermine the rights or interests of the owner of the page that is linked to.
- Suppose, for example, that X sets up a homepage for her site. On the homepage she places some advertisements, from which she hopes to make some money. The homepage also contains links to various subordinate pages, which contain content that X believes consumers would like to see.

Linking, In-lining And Framing

- Y then creates his own Web site, which contains links to X's subordinate pages. The net result is that visitors to Y's site will be able to gain access to X's material, without ever seeing X's advertisements. This type of activity is called "deep linking." (**Intellectual Property in Cyberspace(harvard.edu)**).
- **In-lining** : "In-lining" is the process of displaying a graphic file on one website that originates at another. For example, in-lining occurs if a user at site A can, without leaving site A, view a "cartoon of the day" featured on site B. IMG links -- a special type of link - can be used to display graphic files on one site that are stored on another”.

(Playboy Enterprises v - NYU Law)

Linking, In-lining And Framing

- **Framing** : "Framing" is the process of allowing a user to view the contents of one website while it is framed by information from another site, similar to the "picture-in-picture" feature offered on some televisions. Framing may trigger a dispute under copyright and trademark law theories, because a framed site arguably alters the appearance of the content and creates the impression that its owner endorses or voluntarily chooses to associate with the framer". (**Playboy Enterprises v - NYU Law**).
- **In** Futuredontics Inc. v. Applied Anagramic Inc, 2007, A district court ruled that the addition of the reproduced Web pages within a "frame" by dental website containing contents of other website.

Search Engines And Their Abuse

- Cyberspace is a virtual place with endless possibilities, where Internet offers search engine to search and find data in cyberspace. Search engine is a ‘searchable index of resources available on internet’.
(Sharma, 2015, 525).
- Search engines connects the user with World Wide Web in one place and is a tool which searches online data or content, for example, Google and Yahoo.
- When any website’s keyword is searched the user is tuned with the original page of the websites or the domain name. However, there is possibility of unfairness in this process that can lead to trademark infringement issues.

Regulatory Frame Work- National And International Scenario

- The internet has created a new virtual world and Information Technology Act, 2000 maintains that world in cyberspace by giving protection to various legal challenges and their suitable solution.
- Intellectual Properties such as copyright, trademark, patent, layout and circuit designs are the new members of this virtual world which exists in the cyberspace.
- Therefore, the protection of these rights is as essential as any other right within the cyberspace and with the ever changing times, the demand for protection and remedies is also changing and the need of new and effective law to protect the new inventions is in demand.(Aiswarya etall ,2018).

Regulatory Frame Work- National And International Scenario

Legal Protection in India:

- Making of copies or adaptation of a computer Programme by the lawful possessor of a copy of such computer Programme from such copy in order to utilize the computer Programme for the purpose for which it was supplied or to make back-up copies purely as a temporary protection against loss, destruction, or damage in order only to utilize the computer Programme for the purpose for which it was supplied;
- The doing of any act necessary to obtain information essential for operating inter-operability of an independently created computer Programme with other programmed by a lawful possessor of a computer Programme, if such information is not otherwise readily available;

Regulatory Frame Work- National And International Scenario

Legal Protection in India:

- In the observation, study or test of functioning of the computer Programme in order to determine the ideas and principles, which underline any elements of the Programme while performing such acts necessary for the functions for which the computer Programme was supplied;
- Making of copies or adaptation of the computer Programme from a personally legally obtained copy for non- commercial personal use. The Copyright, can be assigned or transferred or the owner of the work can license specific uses to another person and accordingly specify the gravity of ownership being given to another person.

Regulatory Frame Work- National And International Scenario

- The Copyright expires after 60 years from the end of the calendar year in which the author dies Literary, dramatic, musical or artistic works;
- The Copyright shall subsist until 60 years from the beginning of the calendar year following the year in which the film/sound recording/ photographs/ computer programs is made available or first published as the case may be to the public.
- The act provides economic rights under sec 14 of act to commercially exploit his creation and also grants moral rights as envisaged under Section 57 of the Act which are special rights of the author of the work viz.,

Regulatory Frame Work- National And International Scenario

- Right to claim authorship of the work
- Right to restrain or claim damages in respect of any distortion, mutilation, modification or other act in relation to the said work if such distortion, mutilation, modification or other act would be prejudicial to his honor or reputation (“Right Against Distortion”).
- The moral rights can also be exercised by legal representatives post death of the author. As per the Amendment, the right against distortion is available even after the expiry of the term of copyright.

Regulatory Frame Work- National And International Scenario

- In addition to the Copyright Act, 1957, there is also Copyright Rules, 1958 and the International Copyright Order, 1999. The Copyright Rules contain the rules and regulations and provides various procedures and where, the International Copyright Order is concerned, it deals with the protection of copyright works of nationals of various foreign countries.
- **Patent-** It is governed by the Patents Act 1970; Patents Rules 1972. Section 2(m) of the Patent Act, 1970 provides for the definition of Patent which states that: - “Patent means patent for any invention granted under this Act”.

Regulatory Frame Work- National And International Scenario

Legal Protection in India

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Regulatory Frame Work- National And International Scenario

Legal Protection in India

- **Trademark:**
- Trade Marks Act (TMA), 1999 protects the rights of the trademark owners or business entities for a term of **10 years** from the date of application, renewable every 10 years on payment of the requisite fee.
- Sec 135 of the provides remedy in suits for infringement or for passing off in form of injunctions and damages. Section 103 imposes penalty for applying false trademarks, trade descriptions which shall be punishable with imprisonment for a term not less than six months but which may extend to three years.

Regulatory Frame Work- National And International Scenario

- Further, section 104 of TMA imposes penalty for selling goods or providing services to which false trade mark or false trade description is applied punishable with imprisonment for a term not less than six months which may be extended to three years and with fine which shall not be less than fifty thousand rupees which may be extended to two lakh rupees.
- Offers or exposes goods for sale, puts them on the market, or stocks them for those purposes under the registered trade mark, or offers or supplies services under the registered trade mark; imports or exports goods under the mark; or uses the registered trade mark on business papers or in advertising.

Regulatory Frame Work- National And International Scenario

Legal Protection in India

- **Three types of remedies are available against infringement of IPR**
- Civil Remedies- injunctions, damages, rendition of accounts, expert order, seizure, destruction or forfeiture of infringing goods.
- Criminal remedies- section 63 of the Copyright Act, 1957 deals with Offences of infringement of copyright and Chapter XII of the Trademarks Act, 1999 deals with offences, penalties and procedures pertaining to trademark infringement.
- Administrative Remedies- import/ export of goods including protection of patents, trademarks and copyrights under Indian Customs Act, 1962.

Regulatory Frame Work- National And International Scenario

Legal Protection in India

- The geographical Indications of (Registration and Protection) Act, 1999 for Laws relating to Geographical Indication; Information Technology Act, 2000 deals with electronic records.



Regulatory Frame Work- National And International Scenario

International scenario

- Important treaties which provide international protection to Copyright are:
- Berne Convention for protection of Literary and Artistic Works, 1886.
- Universal Copyright Convention, 1952.
- Agreement on Trade related Aspects of Intellectual Property Rights, 1994.
- The purpose of making these treaties is to create uniformity in dealing with the disputes related to the Intellectual Property Rights, because copyright is governed within the country according to the internal laws of the country and with the help of these treaties, uniform protection is given to all member countries of that particular treaty.

Regulatory Frame Work- National And International Scenario

International scenario:

- The Intergovernmental Copyright Committee has been also established in compliance with Art.11 of the UCC consisting of the representatives of 18 Contracting States. (Universal Copyright Convention. United Nations Educational, Scientific and Cultural Organization (unesco.org))
- **Berne Convention-**
 - The Berne Convention deals with the protection of works and the rights of their authors. It is based on three basic principles: (1) Works originating in one of the Contracting States (that is, works the author of which is a national of such a State or works first published in such a State)

Regulatory Frame Work- National And International Scenario

International scenario

- It must be given the same protection in each of the other Contracting States as the latter grants to the works of its own nationals (**principle of "National treatment"**), (2) Protection must not be conditional upon compliance with any formality (**principle of "Automatic" protection**), (3) Protection is independent of the existence of protection in the country of origin of the work (**principle of "Independence" of protection**).
- If, a Contracting State provides for a longer term of protection than the minimum prescribed by the Convention and the work ceases to be protected in the country of origin, protection may be denied once protection in the country-of-origin ceases.

Important Topics



Summary

- In this session, we discussed the term Intellectual Properties and Cyberspace is entirely different but in digital world almost every information is available over cyberspace and because of this, many of the intellectual property works are getting infringed or being misused which results in consumers being misled and violation of the rights of the owners of the intellectual property.
- The rationale behind providing Intellectual Property rights and legal protection to the creators and inventors is to give them the due recognition for their intellectual work and also the monetary benefits for certain period of time to encourage further innovations; economic and technological growth.

Summary

The IPR violations in digital media like Copyright's violations, Framing, abuse of search engines by use of Meta-tags, spamming and especially trademark violation giving rise to Domain Name Disputes are major concerns.

*Thank
You*